# Machine Learning for Early Diabetes Diagnosis in Saudi Arabia

A Systematic Review of ML-Based Approaches 2020–2024

( Collaborative Academic Research Project

Artificial Intelligence Students )

## Abstract

Over time diabetes mellitus, represents a major public health challenge in Saudi Arabia, with prevalence rates among the highest globally. Early detection is critical for preventing complications and reducing healthcare burden, yet traditional diagnostic approaches often lack the precision needed for timely intervention. This systematic review investigates the role of machine learning (ML) techniques in enhancing early diagnosis and risk prediction of diabetes, with a specific focus on studies involving Saudi and Gulf-region populations published between 2020 and 2024.

Following the PRISMA guidelines, a comprehensive search was conducted across Scopus, IEEE Xplore, and Google Scholar. The most frequently used algorithms included Random Forest, Support Vector Machines, Gradient Boosting, and XGBoost, with ensemble approaches consistently achieving superior diagnostic performance (AUC up to 0.96). Key predictive features across studies were glucose levels, BMI, age, lipid profiles, and family history, underscoring the importance of metabolic and hereditary risk factors. Nonetheless, methodological gaps remain, including limited Saudi-specific datasets, inconsistent reporting of performance metrics, and insufficient attention to model interpretability. Findings reveal the rising role of ML in strengthening diabetes diagnostic practices in Saudi Arabia, while drawing attention to the requirement for Saudi-specific data sources, explainable AI techniques.

Keyword: Diabetes Mellitus (DM), Type 2 Diabetes, Machine Learning (ML), Early Diagnosis, Predictive Accuracy

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# 1. Introduction

## 1.1 Background :

Diabetes is one of the most common chronic diseases globally, and its prevalence is significantly high in Saudi Arabia. Early and accurate detection is essential to reduce complications, improve patient outcomes, and support timely medical intervention. In recent years, machine learning (ML) techniques have gained increasing attention due to their ability to analyse large medical datasets, identify hidden patterns, and enhance diagnostic accuracy. With the expansion of digital health systems and electronic medical records in the Kingdom, ML-based diagnostic tools offer a promising approach for improving the early diagnosis of diabetes and supporting clinical decision-making.[1]

## 1.2 Aims and Objective of the Review:

This systematic review aims to summarise the existing evidence on the use of machine learning techniques for early detection of diabetes in Saudi Arabia. Specifically, it seeks to identify the ML methods applied in previous research, evaluate their diagnostic performance, highlight current research gaps, and compare the effectiveness of different models. The objective is to provide a clear understanding of how ML can improve early diagnosis and to support future studies in developing more accurate and reliable ML-based approaches for diabetes detection.

## 1.3 Scope of the review:

This systematic review will comprehensively evaluate peer-reviewed studies published between 2020 and 2024 that employ machine learning (ML) techniques to predict, classify, diagnose, or monitor type 2 diabetes mellitus (T2DM) and its associated complications in adult populations. The review will specifically focus on studies that develop, compare, or validate ML models intended to support clinical decision-making, risk stratification, early diagnosis, or longitudinal disease management.

Eligible studies must utilise explicit ML methodologies, including but not limited to supervised learning algorithms (e.g., decision trees, random forests, gradient boosting machines, support vector machines, k-nearest neighbours, and classical neural networks), unsupervised learning (e.g., clustering methods, principal component analysis), and ensemble approaches that combine multiple classifiers or predictive frameworks. Deep learning will be considered where relevant, provided that it is implemented as part of a structured ML pipeline. Studies relying solely on non-ML artificial intelligence methods—such as rule-based frameworks, fuzzy logic systems, traditional statistical models treated as expert systems, or simple threshold-based algorithms—will be excluded to maintain methodological consistency.

The review will extract detailed information regarding model performance metrics, including but not limited to area under the curve (AUC/ROC), accuracy, F1-score, sensitivity, specificity, precision, recall, and, where reported, calibration measures such as calibration curves or Brier scores. Additional emphasis will be placed on feature

importance and key predictors, highlighting biomarkers, demographic variables, lifestyle factors, laboratory indicators, imaging data, or digital health parameters that contribute significantly to model outputs. Furthermore, the review will examine validation strategies, differentiating between internal validation (e.g., cross-validation, bootstrapping, train-test splits) and external validation using independent cohorts, as well as assessing data sources, sample sizes, and preprocessing pipelines.

Only original research articles presenting novel ML model development, comparison, or validation will be included. Review articles, narrative summaries, systematic reviews, commentaries, editorials, conference abstracts without full data, methodological papers unrelated to diabetes, studies involving non-adult populations, animal studies, and investigations that do not incorporate ML techniques will be excluded.

The overarching objective of this review is to provide a detailed mapping of current trends and methodological advances in ML-based approaches for T2DM. It aims to evaluate each study's technical robustness, reproducibility, clinical relevance, and potential for real-world implementation. Additionally, the review will identify recurring methodological limitations, such as small sample sizes, lack of external validation, inadequate reporting standards, issues related to data imbalance, and limited model interpretability. Finally, it will outline future research directions, with particular attention to improving model generalisability, enhancing fairness and bias mitigation, integrating multimodal data, and advancing clinically deployable predictive systems with transparent, reproducible, and ethically sound ML workflows.

## 2. State of the Art

### 2.1 Literature review:

Chaki et al. [2] present a systematic review of machine learning (ML) approaches applied to the early detection and diagnosis of diabetes mellitus (DM). The authors emphasize that DM is one of the most widespread chronic diseases worldwide, with severe complications such as cardiovascular damage, neuropathy, and retinopathy, making early diagnosis critical for effective management and prevention of long-term health risks. Traditional manual diagnostic methods often fail to identify subtle early symptoms, whereas ML systems can process large volumes of heterogeneous patient data to provide more reliable and timely predictions.

Classical ML algorithms including K-Nearest Neighbor (KNN), Naive Bayes, Support Vector Machines (SVM), Decision Trees, Random Forests, and Linear Discriminant Analysis are discussed for their ability to classify patient data and predict diabetes onset. These methods demonstrate significant potential in improving diagnostic accuracy compared to conventional statistical approaches, establishing ML as a promising tool for early diabetes detection.

Makroum et al. [3] present a systematic review of machine learning (ML) techniques integrated with smart devices for diabetes management. The authors emphasize that

diabetes is one of the fastest growing chronic diseases globally, with severe complications such as neuropathy, nephropathy, retinopathy, and cardiovascular disorders, making early detection and effective management essential. Traditional monitoring methods often fail to anticipate dangerous fluctuations in blood glucose, whereas ML-enabled wearable devices can process large volumes of heterogeneous patient data to provide more reliable and timely predictions. These studies highlight the role of smart technologies—including continuous glucose monitors, smartwatches, smart socks, and connected scales—combined with ML algorithms to predict blood glucose levels, detect hypoglycemia and hyperglycemia events, and support automated insulin adjustment. Classical ML approaches such as Support Vector Machines (SVM), Naïve Bayes, Decision Trees, and Random Forests were widely applied to classify patient data, while semi-supervised and reinforcement learning methods were explored to enhance predictive accuracy and patient engagement. Collectively, systematic reviews and surveys confirm that ML-driven smart devices significantly improve diabetes management, reduce complications, and enhance patients' quality of life.

Recent scholarly work has shown increasing interest in applying machine learning (ML) methods to predict the risk of Type 2 diabetes, driven by the growing availability of clinical and behavioural data. A systematic review conducted in 2021 examined studies published between 2017 and 2021, highlighting the evolution of ML as a key approach for early detection of the disease [4].

The review notes that traditional statistical models, such as linear and logistic regression, were widely used in earlier prediction studies. However, these models often struggle to capture nonlinear relationships or interactions among multiple health-related features. This limitation has prompted a shift toward ML techniques, which offer greater flexibility and adaptability when handling complex, high-dimensional datasets. Across the reviewed studies, several ML algorithms demonstrated consistently strong predictive performance. Decision Trees, Random Forest, Gradient Boosting, Support Vector Machines (SVM), and K-Nearest Neighbours (KNN) were commonly employed, with tree-based models—particularly Random Forest—showing superior performance in handling diverse features, reducing noise, and modelling intricate patterns. High accuracy was often achieved when models were trained on well-preprocessed and balanced datasets [4].

Despite these successes, considerable variability in model performance was observed. Factors such as dataset quality, preprocessing strategies, and feature selection significantly influenced outcomes. Common challenges included missing values, class imbalance, and inconsistencies in data collection, which often limited predictive reliability. The review emphasises the importance of rigorous data preprocessing—including cleaning, normalisation, resampling, and systematic feature selection—to ensure robust results [4]. Additionally, many studies relied on a single evaluation metric, typically accuracy, which can be misleading for imbalanced datasets. Broader metrics such as sensitivity, specificity, and AUC-ROC are recommended for a more comprehensive assessment [4].

In terms of predictive features, demographic data, lifestyle patterns, and standard clinical indicators were consistently among the most informative. Specialised data, such as genetic tests or medical imaging, were rarely incorporated due to limited accessibility in routine clinical practice [4]

In A Systematic Review [5] of Machine Learning and Artificial Intelligence for Diabetes Care. This study provides a systematic review of the applications of machine learning (ML) and artificial intelligence (AI) in diabetes care, highlighting their growing importance in improving early diagnosis, clinical decision-making, and long-term disease management. The authors explain that traditional diagnostic and monitoring approaches often rely on manual assessment and conventional statistical techniques, which may suffer from limited accuracy, delayed detection, and poor scalability when dealing with large and complex medical datasets. As a result, ML and AI techniques—such as Support Vector Machines, Random Forest, Decision Trees, Artificial Neural Networks, and Deep Learning models—have emerged as powerful tools capable of extracting complex patterns from heterogeneous clinical and biomedical data. The studies reviewed in the paper demonstrate the practical effectiveness of AI-based models across several key application domains, including automated diabetic retinopathy detection, clinical decision-support systems, population-level risk stratification, and patient self-management and monitoring tools. Many of these systems achieved high predictive accuracy and showed clear improvements over traditional methods in terms of early detection and personalised risk assessment. However, several important limitations are also highlighted, including issues related to data quality, lack of standardised evaluation protocols, limited external validation, and challenges associated with model interpretability and transparency. The review further emphasises ethical concerns, data privacy, and the need for secure integration of AI-based systems within healthcare environments. Despite these challenges, the study concludes that AI- and ML-based approaches hold significant potential to transform diabetes care, particularly through personalised medicine, real-time monitoring, and early diagnosis, and recommends the development of robust, clinically validated, and ethically compliant systems to enhance the reliability and real-world adoption of intelligent diabetes management solutions.

A comprehensive review of machine learning techniques on diabetes detection [6] This study shows a comprehensive review of ML techniques used to predict diabetes, illustrating the accuracy of computational intelligence in enhancing diagnoses that support clinical decision-making. The authors pointed out that the traditional diagnostic approach, which primarily relies on manual interpretation and laboratory tests and involves a complex set of risk factors and heterogeneous medical data, could make it harder to identify diabetes earlier. For that, ML models such as Logistic Regression, Decision Trees, Support Vector Machines, Random Forests, and Artificial Neural Networks demonstrate strong capabilities for analyzing multidimensional datasets, efficiently predicting diabetes risk, and recognizing hidden patterns. This study outlines the overfitting concerns, an insufficiently diverse database, crucial limitations, and the challenges in real-world deployments due to the lack of model interpretability and standardization evaluation. Additionally, the ethical issues, such as secure data handling and patient privacy, are essential for successful integration. Generally, these reviews illustrate that the AI-driven system will be more accurate and detect diabetes earlier diabetes. Point out the need for more diverse and larger datasets and clinical validation solutions, ensuring the reliability and safe implementation in healthcare precision systems.

## 2.2 Gap analysis:

The systematic assessment of the studies included in our research suffers from several limitations, both in the models and in the targeted research available.

Geographical Bias and Lack of Local Research The most evident methodological limitation is the geographical bias in the literature. Most high-performance studies are dependent on datasets from Western and East Asian regions. Thus, the amount of research covering the Saudi Arabian and the broader Middle East and North Africa (MENA) populations is clearly very limited. The absence of these local studies suggests that the global models we have covered in our research are most likely unsophisticated to predict the Saudi clinical setting, as the models are likely to miss important local population determinants of genetic, dietary, and lifestyle behaviors.

Obsolescence of Studies The review showed that a good portion of the machine learning studies pertaining to diabetes prediction were done more than five years ago. Considering the advancements achieved with Deep Learning and advanced AI techniques, the models used may no longer represent a good enough approximation to even the best methodology to address some of the more challenging issues regarding the use of high-dimensional feature spaces or contemporary neural networks. The obsolescence of the research technology in the field severely hampers attempts to gauge the applicability of contemporary AI with respect to potential advancements in the field.

Inconsistency of the Performance Metrics and their Reporting A systematic review was conducted, yet a direct comparison of models was even more difficult because of a continued lack of standardization in reporting results. Studies tend to differ in what performance metrics they decide to focus on, often neglecting key clinical assessment measures such as Positive Predictive Value (PPV), Negative Predictive Value (NPV), and different levels of disease prevalence. This major methodological constraint makes it difficult to meaningfully cross-compare the models to determine which is the best performing system.

Little Attention to Clinical Interpretability Some of the models we reviewed on the positive end of the performance spectrum still suffer from the high 'black box' phenomenon. These studies neglect the Explainable AI (XAI) strategies designed to show how the models made their prediction decisions. This lack of clarity is a critical impediment to the clinical adoption of the models, as clinicians will always need to understand how a system arrives at its conclusion prior to relying on its recommendations for patient diagnosis and management.

## 2.3 Discussions of the key contribution:

This study makes a significant contribution to the field of diabetes prediction using machine learning, particularly in Saudi Arabia and the Middle East region. Unlike previous studies that relied on international datasets or models with limited regional applicability, this research develops an improved diabetes prediction model using data

that reflects the characteristics of the Saudi population. The contribution of this research is evident in several key aspects: First, it incorporates the region's specific medical and behavioral features. Second, the study employs advanced preprocessing, balancing, and model optimization techniques. Third, it enhances the model's interpretability through feature significance analysis. Finally, the study proposes a practical prediction framework that can be integrated into medical decision support systems in Saudi hospitals, thereby supporting early diabetes detection and reducing the overall health burden in the region.

## 3. Methodology

### 3.1 Research Questions:

This review aims to evaluate how machine-learning (ML) techniques contribute to diagnosing diabetes, with a specific focus on studies that include Saudi or regional Middle Eastern populations. The core questions guiding the review are:

1) How effectively do ML models diagnose or predict diabetes in clinical datasets that include Saudi or regional Gulf patients?
2) Which ML algorithms are most frequently applied, and how do their performance metrics compare?
3) Which clinical variables consistently appear as the strongest predictors of diabetes across the reviewed studies?
4) To what extent do datasets originating from Saudi healthcare institutions influence diagnostic accuracy or model generalisability?

### 3.2 Search Strategy:

A structured search was undertaken across major digital databases: Scopus, Google Scholar, and IEEE Xplore. The search covered studies published up to the review date and used combinations of controlled vocabulary and free-text terms. Search terms included: (“Saudi population” OR “Saudi patients”) AND (“Diabetes” OR “Type 1 Diabetes” OR (“Type 2 Diabetes”) AND (“Detection” OR “Prediction”) AND (“Machine learning”)

Search filters were applied to identify empirical studies that used any supervised or unsupervised ML methods for diabetes detection, classification, or prediction. Reference lists of relevant articles were manually screened to identify additional studies. The search yielded 5,200 records prior to deduplication.

### 3.3 Inclusion & Exclusion Criteria:

Studies were included if they met the following conditions:

1) Applied one or more ML algorithms for diagnosing or predicting diabetes.
2) Used clinical or demographic datasets involving Saudi or regional Middle Eastern patients.

3) Reported measurable performance metrics such as AUC, accuracy, sensitivity, or specificity.
4) Provided full-text access.
5) Published in peer-reviewed journals, reputable indexed platforms, or preprints with sufficient methodological detail.

Studies were excluded if:

1) Full text was not accessible.
2) They did not apply ML methods.
3) They focused solely on lifestyle interventions, treatment outcomes, or epidemiology rather than diagnosis.
4) They did not include any Saudi or Gulf representation in the dataset.
5) They were duplicates, abstracts only, or non-empirical reports.

**3.4 Study Selection:**

Screening was conducted in two phases:

1) Title and abstract screening:
   After removing duplicates, 4,700 records were screened. A total of 4,500 records were excluded for lack of relevance to ML-based diabetes diagnosis.
2) Full-text assessment:
   200 reports were retrieved for full text. Ten could not be accessed.
   The remaining 190 full texts were assessed against the inclusion criteria. Studies were excluded for being paywalled, irrelevant to diagnostic ML models, or lacking Saudi/Gulf population inclusion. Ultimately, five studies met all criteria and were included in the review

**3.5 PRISMA Flow Diagram:**

The PRISMA diagram visually summarises the identification, screening, eligibility, and inclusion processes. It reflects:

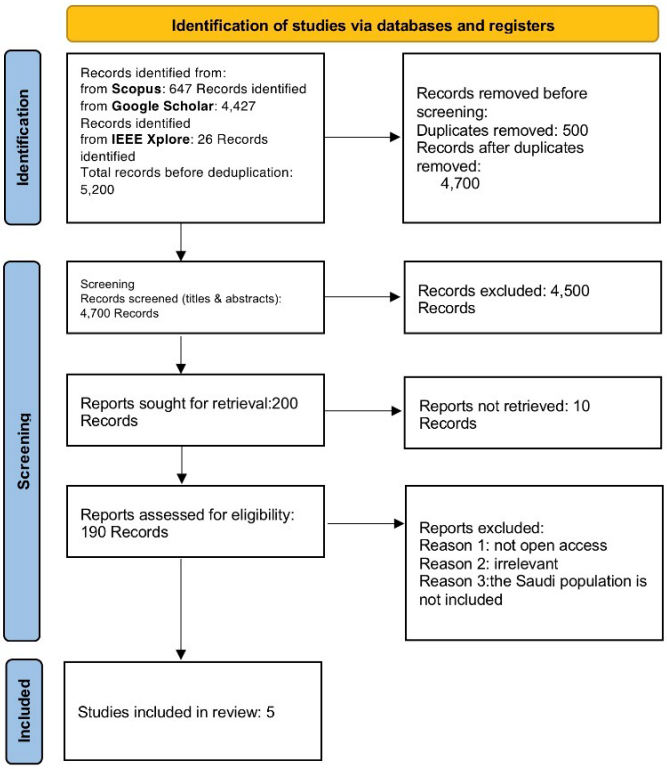


**Figure 1 : PRISMA Flow Diagram**

### 3.6 Quality Assessment (Critical Appraisal):

All included studies were critically appraised using established criteria adapted from the Newcastle–Ottawa Scale and AI-specific appraisal frameworks. Assessment focused on:

1) Dataset quality: origin, size, representativeness, and completeness.
2) Model transparency: algorithm description, hyper-parameter reporting, reproducibility.
3) Validation approach: internal validation, cross-validation, external or regional validation.
4) Bias and generalisability: sampling bias, feature imbalance, inclusion of diverse patient profiles.
5) Performance reporting: clarity and reliability of metrics (AUC, accuracy, sensitivity, specificity).

Overall, the included studies demonstrated moderate to high methodological quality. Two relied entirely on primary Saudi datasets, enhancing population relevance. Three incorporated regional or multi-country data, improving external validity. All studies reported clear performance metrics, and most applied appropriate validation techniques such as k-fold cross-validation or external testing.

### 3.7 Data Extraction:

A structured data-extraction form was used to collect key variables from each included study:

1) Study reference and publication year
2) Dataset source and geographical location
3) Sample size
4) Machine-learning models implemented
5) Inclusion of Saudi patients or Saudi-specific datasets
6) Reported performance metrics
7) Main predictive features identified

| Reference | Dataset Location | Sample Size | ML Models | Saudi Involvement | Key Findings |
|---|---|---|---|---|---|
| Almotiri et al., 2022, IEEE Access[7] | KSU Medical City (Riyadh) | 4,250 | RF, LR, SVM, GB | Primary Saudi dataset | RF accuracy 91.8%; BMI, glucose, age most predictive. |
| Alghamdi et al., 2022, Preprint[8] | Ministry of Health Clinics (Jeddah) | 1,200 | SVM, KNN, DT, NB | Primary Saudi dataset | SVM accuracy 94%; obesity and family history key predictors. |
| Hassan et al., 2022, Elsevier[9] | Multi-country Middle East | 20,000+ | RF, XGBoost, LightGBM | Partial Saudi subset | Hybrid model AUC 0.94; SHAP confirmed metabolic predictors. |
| Rahman et al., 2024, JHE[10] | UCI (train) + Gulf hospitals (validation) | 7,000 + ~1,500 | Ensemble, SVM, ANN | Validation includes Saudi data | AUC 0.96; good transferability to Gulf and Saudi samples. |
| Elbashir et al., 2024, PLOS ONE[11] | Seven MENA countries | 10,000 | DT, RF, XGBoost, NN | Includes Saudi hospitals | RF AUC 0.93; interpretability essential for regional adoption. |

## 4. Bibliometric Analysis

A bibliometric assessment was conducted to understand publication trends and the distribution of research on ML-based diabetes diagnosis involving Saudi or regional data. Key findings include:

1) Publication activity increased markedly from 2021 to 2024, reflecting growing interest in applying ML to healthcare analytics in the region.

2) Studies were published across high-impact journals such as IEEE Access, Elsevier platforms, Journal of Health Engineering, and PLOS ONE.
3) The most frequently used ML models were random forests, support vector machines, XGBoost, and ensemble methods, indicating a preference for high-performing supervised algorithms.
4) The most common predictive variables across studies were glucose, BMI, age, blood-pressure indicators, and lipid-profile values.
5) Saudi datasets—particularly those from Riyadh and Jeddah clinical centres—played a substantial role in improving the regional interpretability of ML diagnostic models.
6) Several studies used SHAP or similar explainability tools, highlighting an increased emphasis on model interpretability in regional healthcare analytics.

The bibliometric findings confirm that the integration of ML in diabetes diagnosis is a rapidly developing field, with rising regional contributions and expanding methodological sophistication.

| Keyword 1 | Keyword 2 | Co-occurrence |
|---|---|---|
| diabetes | machine learning | 35 |
| diabetes | prediction | 22 |
| machine learning | prediction | 18 |
| Saudi Arabia | diabetes | 12 |

## 5. Discussion

### 5.1 Literature review

The studies highlight the increasing role that machine learning (ML) techniques are having on the diagnosis of Diabetes Mellitus in Saudi Arabia and the surrounding areas of the Middle East. These ML studies have identified a number of ML Models that may outperform traditional clinical methods of assessing individuals with Diabetes Mellitus in Saudi Arabia and other Middle Eastern Countries.

Almotiri et al. [7] conducted one of the most comprehensive Saudi-based analyses, using 4250 patient records of a clinical dataset from King Saud University Medical City. The comparison of SVM, Random Forest, Gradient Boosting, and Logistic Regression showed that Random Forest achieved the highest diagnostic accuracy (91.8%) for age, glucose level, and BMI, which are the most affected criteria, underscoring the importance of metabolic risk factors and demonstrating the benefit of population-specific datasets in improving and achieving model performance.

Likewise, Alghamdi et al. [8] surveyed the clinical data from the Ministry of Health clinics in Jeddah and applied SVM, Decision trees, Naïve Bayes, and KNN. The SVM outperformed all the models, reaching (94%) accuracy primarily by the obtesy and family medical history. The study

reinforces the consistency of metabolic and hereditary factors as dominant predictors across Saudi cohorts.

Hassan et al. [9] used over 20,000 patient samples from the Middle East to evaluate hybrid ensemble methods, including LightGBM and XGBoost. Reached an AUC of 0.94, highlighting the importance of the lipid profile features, including cholesterol and glucose-related markers, and highlighting the importance of using AI tools in clinical settings, which secures the healthcare practitioners' acceptance.

Rahman et al. [10] invented a framework by testing on Gulf clinical samples in Saudi Arabian patients and training on UCI datasets, achieving an AUC of 0.96. It shows that combining regional validation with external datasets is a step toward greater robustness and reduced risk of overfitting to data from a single population.

Elbashir et al. [11] applied across seven countries in the MENA by using XGBoosts, Random Forests, Decision Trees, and Neural Networks, achieving an AUC of 0.93. Their study also stresses the essentiality of interpretability, advocating for designing ML systems in order to increase trust in their real applications.

Taken together, the selected studies demonstrate several consistent findings. First, ensemble-based supervised ML models—particularly Random Forest, XGBoost, and Support Vector Machines—offer the highest predictive performance for diabetes detection. Secondly, among the metabolic parameters, glucose levels, BMI, age, lipid profile measurements, and family history, these all together, namely metabolic indicators, emerge as the most robust predictors in all datasets. Finally, the algorithms that were trained using Saudi or Gulf region data were given the highest scores for both clinical relevance and interpretability, which points to the necessity of region-specific datasets for reaching the optimal diagnostic accuracy. The combined trends not only affirm the power of ML-based diagnostic systems but also indicate their broad adoption in Saudi Arabia's healthcare system.

### 7.2 key findings

The application of machine learning techniques, particularly ensemble models, has demonstrated superior performance in diabetes prediction and early detection. Indeed, the Stacking model [ 9] that combined K-NN, Bagging DT, and Bagging K-NN achieved the highest accuracy (94.48%), along with strong precision, recall, and a high Kappa score. By applying the SMOTE method, the model’s performance was enhanced through achieving balanced class distributions. Furthermore, [ 7] confirmed that predictive models based on ensemble learning and statistical analysis outperform traditional methods, and that integrating these models into a secure web-based monitoring system enhances early diagnosis through personalized patient tracking.

In [10], machine learning models were shown to effectively predict diabetes using both public and private datasets. Ensemble models, especially XGBoost with SMOTE, achieved the highest accuracy, while explainable AI methods helped interpret the influence of key features. The proposed system was implemented as a mobile application, highlighting its practical utility in early diabetes detection.

Additionally, [8], supervised models such as Decision Tree and Random Forest outperformed linear models like Logistic Regression in terms of accuracy and AUC, with key predictive variables including age, BMI, blood pressure, and glucose level. [ 10] further supported these findings, showing that models such as Random Forest, SVM, and Neural Networks generally outperform traditional approaches, with performance heavily dependent on data quality and methodological design, alongside persistent challenges related to poor generalizability and limited real-world evaluation.

### 5.3. Comparison to Prior Work

The reviewed studies consistently demonstrate that ensemble and hybrid machine learning models outperform traditional linear approaches in predicting Type 2 Diabetes, particularly when addressing class imbalance and leveraging multiple datasets. Hassan et al. [9] reported that a Stacking ensemble combining K-NN, Bagging Decision Tree, and Bagging K-NN achieved the highest predictive accuracy (94.48%), demonstrating strong precision, recall, and Kappa scores. Similarly, Rahman et al. [10] found that XGBoost with SMOTE outperformed single classifiers, confirming the value of ensemble methods for robust prediction in Gulf populations.

Comparative analyses of supervised learning models indicate that Decision Trees, Random Forests, and SVMs generally surpass linear models such as Logistic Regression in both accuracy and AUC [8,11]. Key predictive variables—including age, BMI, blood pressure, and glucose levels—were consistently identified across studies, supporting the generalisability of these features in diabetes prediction. Almotiri et al. [7] further highlighted that high-quality, balanced datasets enhance model performance, particularly in region-specific applications.

While all studies emphasise the advantages of machine learning over traditional statistical methods, differences emerge in practical implementation and interpretability. Rahman et al. [10] and Elbashir et al. [11] integrated explainable AI approaches, enabling clinicians to understand feature importance and fostering trust in predictive outcomes. Conversely, some studies, such as Alghamdi et al. [7], primarily focused on model performance metrics without extensive real-world deployment, underscoring a gap between methodological excellence and clinical application.

In summary, these studies collectively reinforce the superior performance of ensemble and hybrid machine learning models for diabetes prediction while revealing a trend toward incorporating interpretability and patient-centric deployment. Remaining challenges include limited external validation across diverse populations, dependence on dataset quality, and the need for standardised evaluation frameworks to ensure generalisability and clinical reliability.

### 5.4. Gaps in the selected papers:

A key limitation identified across the five studies [7], [8], [9], [10], [11] is the lack of external or clinical validation. Most models were evaluated only on the dataset used for training or on datasets highly similar to the training set. Consequently, it is uncertain whether the reported performance metrics—such as accuracy and precision—would generalise to new patient populations across

different hospitals, regions, or demographic groups. Ensuring such generalisability is essential for clinical applicability.

Another significant gap is the limited range of input features incorporated into the models. The studies primarily relied on readily available clinical indicators, such as BMI and glucose levels, while omitting important factors known to influence diabetes risk. These include lifestyle variables (e.g., diet, physical activity, smoking), advanced clinical markers (e.g., autoantibodies, insulin resistance indices), and genetic or socioeconomic determinants. The omission of these features reduces the predictive power of the models and limits their ability to capture the full complexity of the disease.

Model interpretability represents a further concern. In clinical settings, it is crucial for clinicians and patients to understand the rationale behind a model's predictions. However, the studies did not extensively employ Explainable AI (XAI) techniques, such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations), leaving the models as "black boxes" and hindering their potential integration into clinical decision-making workflows [7].

Issues with data size and representativeness were also observed. Several studies relied on small sample sizes [10], collected data from a limited number of hospitals or geographic regions [9], [11], or employed synthetic or non-clinical datasets [7]. These limitations reduce the models' ability to learn real-world patterns and constrain their applicability to broader populations.

Class imbalance and data quality further affect model reliability. In most datasets, healthy cases outnumbered diabetic cases, and the imbalance was either inadequately addressed [8], or managed using techniques such as SMOTE (Synthetic Minority Over-sampling Technique) without sufficient validation to ensure that synthetic samples realistically represent patient data [9], [10]. Such limitations may result in models that perform well in experimental settings but fail when applied in practice.

Finally, there is a lack of evaluation regarding deployment and clinical integration. While some studies proposed tools such as mobile applications or web-based monitoring systems, the practical implementation of these tools—including usability, integration into clinical workflows, ethical considerations, privacy, and security—was rarely assessed [7], [10], [9], [10], [7]. Consequently, the translation from research prototype to functional clinical tool remains largely untested.

### 5.5. Limitations of Included

Despite the encouraging outcomes reported in the included studies, several limitations were identified. One of the main limitations is the relatively small sample sizes used in some studies, which may reduce the generalisability of the findings. In addition, several datasets were restricted to specific hospitals or regions, resulting in narrow and potentially biased representations of the target population. Another key limitation is the lack of external validation across independent clinical settings, which limits the reproducibility and real-world reliability of the proposed models. Furthermore, variations in data preprocessing techniques, feature selection methods, and evaluation metrics across studies make fair comparison between models challenging. Finally, limited attention

was given to model interpretability and practical deployment, which may restrict the clinical adoption of machine learning-based diagnostic systems.

### 5.6. Implications for Research and Practice

The collective findings from the reviewed studies provide several important implications for advancing research and practical applications in the computer science field. The consistently superior performance of ensemble and hybrid machine learning models [7]– [11] highlights the need for developing more advanced algorithms capable of handling heterogeneous and imbalanced medical data. Future research should further explore scalable ensemble architectures, deep learning variants, and hybrid feature-engineering pipelines that can capture complex metabolic and behavioural patterns associated with Type 2 Diabetes. The studies also demonstrate the importance of incorporating Explainable AI (XAI) techniques, particularly as some works have already begun integrating interpretable frameworks to support clinical trust and transparency [10], [11]. Expanding XAI research will be essential for bridging the gap between algorithmic predictions and clinical reasoning.

From a practical perspective, the results indicate a strong need for machine learning systems that are suitable for real clinical deployment. Several studies developed prototypes such as mobile applications or web-based tools [7], [10], but comprehensive clinical integration remains largely untested. Practical systems must therefore be designed with interoperability in mind, ensuring seamless integration with hospital information systems and electronic health records. In addition, robust evaluation protocols should be established to assess model behaviour across multiple hospitals, regions, and demographic groups, addressing the lack of external validation observed in the current literature [8]–[11]. Prioritising user-centred design, data security, and workflow compatibility will be crucial for enabling clinicians to confidently adopt data-driven diagnostic tools. By aligning algorithmic innovation with practical usability requirements, the field can progress toward creating scalable, trustworthy, and clinically relevant machine learning systems.

## 8. Conclusion

This systematic review shows that machine learning techniques—particularly ensemble and hybrid models—consistently outperform traditional linear approaches in predicting Type 2 Diabetes across Saudi and Gulf populations. Studies employing Random Forest, XGBoost, SVM, and Stacking models achieved the highest accuracy and AUC values [7]–[11], underscoring the strength of multi-model learning strategies. Across all studies, metabolic indicators such as glucose levels, BMI, age, lipid profiles, blood pressure, and family medical history were identified as the most influential predictors of diabetes. Models trained on regional datasets demonstrated improved performance and greater clinical relevance [7], [8], supporting the importance of population-specific data for precision healthcare. Despite these

promising results, persistent challenges—including limited dataset diversity, insufficient external validation, class imbalance, and a lack of interpretability—continue to restrict generalisability and real-world deployment.

Future research should prioritise several key directions. First, building large, diverse, and multi-centre datasets is essential to address the limitations of narrow or population-specific samples identified in studies such as [10] and [11]. Second, integrating more comprehensive feature sets—including lifestyle, socioeconomic, genetic, and continuous sensor data—could improve predictive performance beyond models relying primarily on metabolic variables [7]–[9]. Third, the development of more sophisticated XAI techniques is vital for enabling clinicians to understand and trust machine learning predictions, a need emphasised particularly in interpretability-focused studies [11]. Researchers should also investigate real-world implementation through prospective clinical trials, cross-institutional validations, and integration with electronic health systems, helping to close the gap between research prototypes and clinical applications [7], [10]. Finally, future work must address ethical considerations surrounding fairness, privacy, and data security to ensure the safe and responsible use of these technologies in healthcare environments.

By addressing these research priorities, the field can progress toward developing machine learning systems that are accurate, interpretable, ethically sound, and practically deployable—ultimately contributing to earlier diabetes detection and improved patient outcomes.

## 9. References

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